Index Term	Links
4-quadrant drives	144
A/D chatter	188
Abnormal Situation Management®	
(ASM®) Consortium	260
AC motors	
multiple pole	141
standard AC induction	140
stepper	141
synchronous	141
types	140
vector	141
wound rotor	141
AC outputs	110
AC versus DC	142
access control	349
accessories	56
accuracy	3
active tags	113
ActiveX	353
actuation	104
electrical	107
actuators	56
electric	56
hydraulic	56
pneumatic	56
adaptive control	179
adaptive motor model	148
adaptive-tuning	40
advanced process control	175
costs and benefits	189
advanced regulatory control	41
alarms	96 123 229–230 355 360
clarity	235
design	231
display	356
floods	234
generator	356
logger configuration	357
management	229
monitoring	233
nuisance	234
philosophy	229
rationalization	230
reports	357

Index Term	Links	
safety	235	
stale	234	
summary display	357	
suppression	232	
testing	404	
training	232	
alternating current	136	
ambient temperature	134	
amiable	476	
analog		
communications	61	
inputs	104	403
output	403	
analytical	476	
analytical instruments	19	
installation	22–23	
maintenance	24	27
analyzer		
hardware	25	
shelters	25	
types	20	
annunciation	247	
ANSI/ISA S50.1-1975	63	
ANSI/ISA-50.00.01-1975	63	
ANSI/ISA-75.01.01-2002		
(60534-2-1 Mod)	53	
ANSI/ISA-88	193	
ANSI/ISA-95.00.03-2005	333	
application requirements	121	
application servers	98	
approximations	240	
area classification	284	
armature	133	
artificial neural network	168	
AS-i	92	
ASME/ANSI B16.34-1996	52	
assessing project status	470	
asset measures	431	
asynchronous	352	
audit	349	
audit trail	98	
authentication	349	
authorization	349	
automatic process control	29	
automation	150	
benefits	441	

Index Term	Links			
engineer	476			
leader	476			
level software	379			
project manager	476			
automation software	373			
history	375			
automation systems	342	442		
auto-tuning	40			
availability	237	241	380–381	424
steady state	242			
backlash	188			
backup	95			
BACnet	220			
ball valves	49			
base	108			
basic process control system (BPCS)	253			
batch process	191			
configurations	192			
versus continuous process control	192			
bellows	4			
bench calibration	403			
block valve	403			
blocking	349			
bourdon tubes	4			
braking methods	145	148		
bridge	145			
brushes	133			
bubbler	6			
buffer	270			
building automation	217			
future of web services	222			
open protocols	220			
specifying systems	222			
bumpless transfer	34			
business planning	335			
butterfly valves	49–50			
bypass	151			
cage-guided valves	51			
calibration	26	403		
capacitance	6			
capital expenditure	375			
capital projects	442			
cascade control	42–43			
cavitation	54–55			

Index Term	Links	
change control	98	
chopper	150	
clean	270	
climate control systems	217	
closed loop tests	38	
coasting	144	
Code of Federal Regulations	203	
collector	108	
commissioning	256	
communication	474	
commutator	133	
compliance	380	
compound wound	136	
conceptual design	256	
configuration	95	
conflict resolution	477	
connectivity	98	346
consensus standards	87	
constant failure rate	239	
constraints	457	
contactors	270	
contingency	460	
continuous control	29	
characteristics	31	
continuous emission monitoring system	203	
contracts	456	
control		
algorithms	93	
cascade	42	
equipment	89	483
decoupling	45	
feedforward	43	
logic	29	
network	93	
objectives	121	
override	46	
ratio	41	
recipe	199	
selector	46	
system parameters	122	
valve	49	
control modules	93	197
control algorithms	93	

Index Term	Links	
controlled variable output processing	94	
diagnostics	94	
discrete input variable processing	94	
historian/trending	94	
process variable processing	93	
sequential control	94	
system access control enforcement	94	
controlled variable output processing	94	
controller	30	158
output	293	
tuning	36	
conversations	475	
converter	145	
cooling	138	
cooling methods	134	
coriolis	4	
cost driven	457	
cost measures	432	
cost-plus	458	
CtrlSpecBuilder	222	
current controller	143	
current source inverter	147	
custom graphic displays	96	
customer engineers	422	
customer service engineers	422	
customer service representatives	422	
dangerous failures	258	
data compression algorithms	367	
data documentation	371	
data management	363	
data quality	370	
data relationships		
entities	363	
many-to-many	364	
one-to-many	363	
one-to-one	363	
data security	372	
data servers	380	
data validation	349	
database	94	351
design	365	
maintenance	372	
operations	367	369

Index Term	Links		
software	370		
database structure	363		
fields	363		
key field	363		
records	363		
DC injection braking	149		
DC motor technologies	107		
DC motors	133		
compound wound	136		
parallel (shunt) wound	135		
permanent magnet	136		
series wound	135		
types	135		
DC outputs	107	109	
DCOM setup	377		
DCS evolution	99		
DDE client	353		
deadband	4	186	232
decay ratio	40		
decommissioning	256		
decoupling control	45		
definition	462		
defuzzification	178		
deployment	465		
derivative mode	34		
derivative-on-error	35		
derivative-on-measurement	35		
design schedule	469		
detection	349		
device performance	426		
DeviceNet	92		
diagnostics	94		
systems	26		
differential pressure transmitter	7		
digital bus	92		
digital inputs	104		
direct acting	34		
direct current	133		
direct digital control	218		
direct OPC	351		
direct torque control	148		
dirty	270		
discrete control	29	117	
discrete I/O	91		

Index Term	Links	
discrete input	403	
variable processing	94	
discrete output	403	
processing	94	
displacer	6	
distillation	20	
distributed control system (DCS)	30	
documentation	75	
installation details	82	
instrument lists	78	
instrument location drawings	81	
location plans	81	
logic diagrams	81	
loop diagrams	83	
operating instructions	87	
piping and instrument diagrams (P&IDs)	78	
process flow diagram	77	
specification forms	78	
standards and regulations	87	
types	76	
double-acting	106	
drain	108	
drip-proof - fully guarded	134	
drive configurations	151	
driver	476	
dump	270	
dynamic compensation	43	
dynamic data exchange (DDE)	351	
dynamic models	170	173
dynamic prioritization	232	
eccentric disc	51	
electric actuator	56	
electric motors	157	
electrical		
distribution configuration	274	
equipment	267	
installations	263	
motors	104	
noise	109	
safety	407	
electronic		
AC	145	
analog transmission	63	
DC	143	

Index Term	Links	
identification	113	
records	381	
signatures	381	
electrostatic discharge (ESD)	109	
e-mail etiquette	475	
emergency shutdown systems	99	
emitter	108	
emulation	98	
encapsulation	289	
enclosure types	134	138
encoders	104	270
encryption	349	
energy management	218	
enterprise asset management	423	
entities	363	
entity-relationship diagrams	364	
environmental conditioning	22	
environmental control		
accuracy	205	
issues	205	
maintenance	205	
environmental controls	203	
Environmental Protection Agency	203	
equipment	336	
entities	194	
malfunction	428	
modularity	194	
modules	198	
ergonomics	359	
estimate	468	
ethernet	92	
execution time	170	
explosionproof	288	
expressive	476	
eXtensible Markup Language (XML)	374	
external feedback	47	
external reset	47	
extrapolation	171	
extreme temperature	408	
faceplates	96	
factory acceptance testing (FAT)	405	
documentation	406	
hardware	407	
operator interface	406	

Index Term	Links		
software	406		
fail-danger	247		
fail-safe	247		
failure modes	245		
annunciation	247		
detected	247		
fail-danger	247		
fail-safe	247		
no effect	247		
undetected	247		
failure rate	424		
feedback	154	160	270
feedback control	31		
loops	29-30		
feedback selection	144		
feedforward control	43		
fiber optic communications	150		
field	133		
field calibration	403		
field devices	257		
final elements	258		
sensors	257		
field effect transistor	108		
field engineers	422		
field exciter	133		
field replaceable units (FRUs)	426		
fieldbus	92		
communications	150		
fields	363		
filter	270		
filtering	349		
final elements	258		
first principal models	169		
steady state	173		
first-order-plus-dead-time (FOPDT)	36		
first-out (first-up)	232		
fixed-price	459–460		
flameproof	288		
flash	54		
flashing	54–55		
flat file	364		
float	6		
floods	97		
flow	8–9		

Index Term	Links
inferential methods	9
velocity methods	10
flowmeters magnetic	
magnetic	10
mass	13
positive displacement	14
turbine	12
ultrasonic	12
vortex shedding	12
fluid power actuation	
control	105
fluid power devices	104
flux vector	147
drive	148
Foundation fieldbus	92
function	193
function block	126
function block diagram	129
fuzzy logic	175
gap analysis	347
gas chromatographs	22
gate	108
gated on	143
GC analyzer oven	25
general purpose requirements	284
general recipe	199
globe valves	49
graphic building	98
graphical user interface (GUI)	96
graphics configuration	352
grounding	264
hall-effect sensors	112
HART	92
hazard analysis	252
heater elements	138
high fidelity	165
historian/trending	94 97
historical data collection	353
HMI design	232
HMI-SCADA systems	99
horsepower	138
human engineering	359
human measures	433
human-machine interface (HMI)	96 123–124 351 399
hybrid	460

Index Term	Links
hydraulic	270
hydraulic actuator	56
HyperText Markup Language (HTML)	374
HyperText Transfer Protocol (HTTP)	374
hysteresis	4
I/O (input/output)	255
IEC 61131	126
IEC 61131-3	129
imperfect testing	244
increased safety	289
indicator lamp	270
industrial networks	313
information technology	218
input/output (I/O)	90
installation	256
details	82
verification	401
instruction list	128
instrument	
commissioning	398–399
engineer	476
index	78
lists	78
location drawings	81
selection	21
instrumentation and control (I&C)	
systems	30
integral mode	33
integration	195
intelligent device management (IDM)	426
interactive PID algorithm	33
interactive screens	366
interfacing	111–112
interlock testing	404
interlocks	123
internal rate of return	445
interpersonal skills	473
interpolation	160 171
interviews	476
intrinsic safety	288
inventory	346
Inverter	270
ISA-5.1-1984 (R1992)	30
ISA-TR84.00.02-2002	258
key field	363

Index Term	Links
keyboard	96
keyseat	140
keyway	140
ladder diagram	128
lambda tuning	38
laminations	136
layer of protection analysis (LOPA)	254
leader/follower	160
leadership	478
leak detection and repair	203
leakage	112
learning process	
motivation	392
reinforcement	392
retention	392
transference	392
level	5
bubbler	6
capacitance	6
differential pressure transmitter	7
displacer	6
float	6
measurement	6
radar (microwave)	6
radioactive	6
ultrasonic	6
level 3-4 boundary	339
level switches	8
paddle wheel	8
tuning fork	8
lifecycle costs	445
lifecycle economics	447
lighting	359
limit switches	57 104 111–112 270
linear	49
dynamic estimator	166 171
motion	51
potentiometers	104
linearity	3
live data	
access	351
interface	377
trend	355
live zero	63
location plans	81

Index Term	Links
logic diagrams	81
logic systems	257
logistics	335
LonMark International	222
loop checks	401–402
loop diagrams	83
loop numbering	78
loop performance	
criteria	36
loop tuning	405
lump-sum	459–460
machine vision	114
magnetic flowmeters	10
magnetostrictive sensing	113
magnetostrictive transducers	104
main processing unit	124
maintainability	421 424
maintenance	256 421 425
management of change	87 229–230 234 471
mandatory standards	87
manometers	4
manual-automatic switching	34
manufacturing execution system (MES)	333
many-to-many	364
mapping	346
mass flowmeters	13
master recipe	199
mean time between failures (MTBF)	241 424
mean time to failure (MTTF)	239
mean time to restore (MTTR)	241
measurement	3 349
flow	8–9
level	5
pressure	4
smart instrument	16
temperature	14
MES	336
integration	335
metal-clad	270
metal-oxide varistors (MOVs)	109
migration	378
mimic graphics	352
mission time	238
mitigation layers	253
model predictive control	180

Index Term	Links	
modifications	256	
monitoring	349	
motion control	153	
motivation	478	
motor power	270	
motoring	145	
MPC models	181	
multiple axes	160	
multiple pole	141	
multivariate statistical process control	167	171
nameplate ratings	135	
National Electrical Code (NEC)	285	
negotiations	477	
net present value	444	
network security	341	
network security techniques	348	
networked I/O	114	
neural networks	168	
Niagara	222	
no effect	247	
normal form	365	
nuisance alarms	234	
oBIX	222	
object linking and execution (OLE)	98	
offset voltage	112	
OLE for process control (OPC)	98	
one-to-many	363	
one-to-one	363	
online version upgrade	95	
on-off controller	39	
OPC	98	
communication	352	
marshalling	378	
status	352	
OPC-Data eXchange (DX)	374	
OPC-XML-DA client	353	
open drip-proof motor	138	
open loop	144	
tests	36	
open-collector	112	
outputs	108	
transistor	108	
operating instructions	87	
operating systems	349	

Index Term	Links		
operational expenditure	375–376		
operational security	341		
operations	256		
operator station	96		
operator training	391		
oscillation	40		
output interfacing	107		
override control (see also selector control)	46		
over-temperature conditions	135		
paddle wheel	8		
parallel (shunt) wound	135		
parallel PID algorithm	33		
partial least squares	168		
passive tags	113		
pattern recognition controllers	187		
people	475		
performance	161		
performance monitoring	26		
period	40		
permanent magnet	136		
PFDavg	248		
pH analyzer	19		
photo electric cell	270		
photoelectric sensors	112		
PID algorithm	32	188	
interactive	33		
parallel	33		
piezoconstrictive devices	104		
piping and instrument diagrams (P&IDs)	30	78	399
PLC	270		
plug stopping	144		
plumbing	352		
pneumatic	270		
pneumatic actuator	56		
pneumatic component interface	91		
pneumatic transmission	62		
position	35		
positioner	56		
position-sensing technologies	112		
positive displacement meters	14		
post-guided valves	51		
potentiometer	113		

Index Term	Links	
potentiometric sensing	113	
precision positioning	57	
presentations	474	
pressure	4	
transducers	57	
transmitters	5	
pressurization	287	
pressurized	288	
pressurized systems	407	
preventive maintenance	428	430
preventive measures	433	
primary controller	42	
principal component analysis	167	
probability	284	
probability of failure on demand (PFD)	248	
probability of failure safely (PFS)	248	
probability of success	237	
process and instrumentation diagrams (P&IDs)	65	
process		
cell	195	
control systems	30	
dead time	186	
flow diagram	77	
heater	46	
process models	165	
costs and benefits	174	
process time constant	186	
variable processing	93	
visualization	351	
product definition	337	
management	338	
production capability	337	
data collection	338	
dispatching	337	
execution management	338	
losses	428	
operations management	336	
performance	336	
performance analysis	339	
resource management	338	
schedules	336	
scheduling	337	
tracking	338	
profibus	92	

Index Term	Links
profit	460
program	125
program code testing	404
programmable logic controller (PLC)	30 99 117-118
programming	150 378
programming language	125
project	455
justifications	441
project management	455
techniques	470
tools	467
project manager	456
proportional band (PB)	33
proportional mode	33–34
proportional-integral-derivative (PID)	
control algorithm	32
proportional-on-error	35
proportional-on-measurement	35
proportionate control valves	107
protection	134 138
protective layers	253
proximity switches	112 270
pseudo random binary sequence	182
pulse counts	93
pulse width modulation	145
purging	287
pushbuttons	270
PWM	270
query	365
radar (microwave)	6
radio frequency identification (RFID)	113
radioactive	6
ratings	134 139
ratio control	41
rationalization	230
real time optimization	183
real-time cost accounting	450
real-time process databases	369
receipt verification	400
recipe	193–194 196 198
control	199
general	199
linkage	199
master	199

Index Term	Links		
site	199		
records	363		
redundancy	95	248	
regeneration	145		
regenerative braking	145	150	
registration	270		
regulators	61		
relational database	364		
relative gain analysis	45		
relay ladder logic	126		
relay method	39		
reliability	237–238	424	
remote accessibility	98		
remote I/O	114		
repairable systems	241		
repeatability	3		
repelling	133		
replacement automation	450		
reports	357	366	474
alarm	357		
request for proposal (RFP)	450		
reset rate	33		
resistive temperature detectors (RTDs)	16		
resolution	188		
resolution limit	188		
resolver	270		
response speed	4		
response time	426		
restoration	244		
restore time	426		
return on investment	443		
reverse acting	34		
risk assessment	252	347	
risk goals	346		
risk reduction	203		
rotary motion	49	54	
rotate	137		
RTD input	403		
RTDs (see resistive temperature detectors)	16		
safe failures	258		
Safety	253	407	
safety functions	253		
safety instrumented systems (SIS)	251		
design life cycle	252		

Index Term	Links	
safety integrated function (SIF)	247	
safety requirements specification	255	
sample conditioning	21	
sample point selection	20	
sampling systems	22	
schedule	199	
scheduled tuning	41	
scope of work	467	
screen management	353	
scripts	358	
secondary loop	42	
security	341	349
plans	344	
policies	344	
programs	344–345	
testing	404	
selector control (see also override control)	46	
self-actuating controllers	61	
self-powered	63	
self-tuning	40	
controllers	179	
semantics	352	
sensing technologies	104	111
sensorless flux vector	148	
sensorless vector	147	
sensors	257	
separation of control	103	
sequence of events	97	
sequential control	94	117
sequential function chart	127	
elements	127	
serial communication	92	150
series wound	135	
server filtering	356	
service level agreements (SLA)	425	
service technicians	422	
serviceability	421	
servo	159	
setpoint	39	
setup	376	
sick building syndrome	205	
silicon controlled rectifier	34	
simulation	98	
single drives	144	

Index Term	Links
single-acting	105
sinking	109
site acceptance testing (SAT)	407
site recipe	199
smart cameras	114
smart instrument	16
SOAP	99
software	125
architecture	375
configuration	377
development	464
fevelopment	404
hierarchy	374
license	377
testing	404
solar control	218
solenoid valve	34 105 108
configurations	105
solenoids	270
source	108
sourcing	109
span	5
specification forms	78
speed	133 138 143 147
control	148
controller	143
reference	143
squirrel-cage motors	138
stakeholders	480
stale alarms	234
standard AC induction	140
standard electronic analog I/O	91
standards and regulations	87
starting torque	141
state-based alarming	232
status report	469
steady state	242
first principal models	173
models	169
stepper	141
stick-slip	188
stiction	4
storage and retrieval	366
structured text	128

Index Term	Links
subscription	352
success triangle	471
support	466
supportability	421
synchronous	141 352
DC-excited	141
non-excited	141
syntax	126
system	
access control enforcement	94
analysis	258
design	463
integration	378
level testing	404
networking	376
outage	124
safety	252
security	341
software	131
workstation	97
tachometer generator	144
TCP/IP	222
teamwork	479
temperature	14 286
temporary relocatable process	
analyzers (TURPAs)	21
testing imperfect	244
testing engineer	476
thermocouple	15 93
third party	41
three phase	270
time and material	459
time and material/not-to-exceed	459
time driven	457
time proportioning	33
torque	134 138 143 147
control loop	148
controlled	147
totally enclosed fan cooled	134 138
non-ventilated	134 138
trainers	393
training	391
evaluation	396
preparation	393
process	392

Index Term	Links
training styles	394
classroom style	395
computer-based training	395
dynamic simulation	395
on-the-job training	395
operating manuals	394
transformer	270
transistors	108
transmitter input	293
transmitters	3 7
trend	353
trend chart viewer configuration	355
trend display	354
trend logger configuration	353
trial-and-error	36
tuning	40
troubleshooting	379 409
circle the wagons method	417
complex-to-simple method	418
consultation	418
DCOM	379
fault insertion method	416
intuition	418
OPC	379
other methods	416
out-of-the-box thinking	418
remove and conquer method	417
seven-step procedure	411
substitution method	416
trapping	417
tuning controller	36
initial	41
scheduled	41
self	40
trial-and-error	40
tuning fork	8
tuning parameters	33
integral mode	33
turbine meters	12
turndown ratio	4
two quadrant operation	144
ultrasonic	6
ultrasonic flowmeters	12
unavailability	242–243

Index Term	Links		
units	194–196		
user system characteristics	121		
user system description	121		
validation	256		
valve deadband	188		
valves ball	49		
butterfly	49–50		
cage-guided	51		
globe	49		
operation	55		
post-guided	51		
selection	53		
types	49		
variable speed drives	143	145	
variable voltage inverter	147		
vector	141		
velocity	35		
vena contracta pressure	54		
vendor assistance	416		
vendor calibration	403		
very dirty	270		
Visual Basic for Applications (VBA)	374		
voicecoil actuators	104		
vortex shedding flowmeters	12		
web services	99		
wet testing	405		
wireless	93		
worm	168		
wound rotor	141		
XML	99		
years of expected life (YEL)	448		
zero calibration	5		
Ziegler-Nichols	36	38	180
zone designation	285		